direction of communication, which provides a useful protection against *reflection attacks*. Each party can contribute a nonce to the KDF, as well as identity information and any other contextual information. For example, if your API can receive many different types of request you could include the request type in the context to ensure that different keys are used for different types of requests.

> **DEFINITION** A ***reflection attack*** occurs when an attacker intercepts a message from Alice to Bob and replays that message back to Alice. If symmetric message authentication is used, Alice may be unable to distinguish this from a genuine message from Bob. Using distinct keys for messages from Alice to Bob than messages from Bob to Alice prevents these attacks.

Table 12.4 COSE HKDF context fields

Field	Purpose
PartyU identity	An identifier for party U and V. This might be a username or domain name or some other application-specific identifier.
PartyV identity	
PartyU nonce	Nonces contributed by either or both parties. These can be arbitrary random byte arrays or integers. Although these could be simple counters it's best to generate them randomly in most cases.
PartyV nonce	
PartyU other	Any application-specific additional context information that should be included in the key derivation.
PartyV other	

HKDF context fields can either be explicitly communicated as part of the message, or they can be agreed on by parties ahead of time and be included in the KDF computation without being included in the message. If a random nonce is